

ALGO ANALYSIS. Correctness Key is in range of the array
 $T(n) = O(f(n))$ upper bound. Performance $A[\text{begin} \dots \text{end}]$
 $\Leftrightarrow \exists c, n_0 (c > 0 \wedge n_0 > 0 \wedge \forall n > n_0 (T(n) \leq c f(n)))$
 For some large enough n , constant multiple of $f \geq T$.
 $T(n) = \Omega(f(n))$ lower bound.
 $\Leftrightarrow \exists c, n_0 (c > 0 \wedge n > n_0 \wedge \forall n > n_0 (T(n) \geq c f(n)))$
 For some large enough n , constant multiple of $f \leq T$.

INVARIANTS.
 Array is sorted. If element in array, $A[\text{begin}] = \text{key}$.
 Array of size n
 Precondition Postcondition.
BINARY SEARCH.

FINDPEAK. Postcondition $T(n) = T(n/2) + O(1)$
 Precondition $A[\text{result}]$ is a local maximum.
 Array of size n $A[i-1] \leq A[i] \wedge A[i] \leq A[i+1]$

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int search(A, key, n)
begin = 0
end = n-1
while begin < end do:
  mid = begin + (end-begin)/2;
  if key <= A[mid] then
    end = mid
  else begin = mid+1
return (A[begin]==key) ? begin : -1
  
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FindPeak(A, n) $T(n) = T(n/2) + O(1)$
 if $A[n/2]$ is a peak then return $n/2$
 else if $A[n/2+1] > A[n/2]$ then
 FindPeak($A[n/2+1..n]$, $n/2$)
 else if $A[n/2-1] > A[n/2]$ then
 FindPeak($A[1..n/2-1]$, $n/2$)
Invariants $O(\log n)$.
 - $\exists$ a peak in the range $A[\text{begin} \dots \text{end}]$
 - Every peak in $A[\text{begin} \dots \text{end}]$ is a peak in $A[0 \dots n-1]$.
 - If we recurse in the right half, then every peak in the right half is a peak in the array.

For some large enough n , constant multiple of $f \leq T$.
Note: $f(n) = O(g(n)) \Leftrightarrow g(n) = \Omega(f(n))$.

$a_k = \alpha_1 r_1^k + \dots + \alpha_k r_k^k$
 Distinct k roots
 $a_k = (\alpha_{10} + \alpha_{11}n + \dots + \alpha_{1m}n^{m-1})r_1^n + \dots + (\alpha_{k0} + \alpha_{k1}n + \dots + \alpha_{km}n^{m-1})r_k^n$
 Non-distinct roots. (distinct roots of multiplicity m_i)
 $x^n = c_1 x^{n-1} + \dots + c_k x^{n-k}$
 Find characteristic polynomial
 $T(n) = c_1 T(n-1) + \dots + c_k T(n-k)$
LINEAR HOMOGENEOUS Recurrences.

Variables: Steep Peak Finding $O(n+m)$
 2D Peak Finding $O(n^2)$
 ① Find MAX on border & cross
 ② If peak, done. Else, recurse on quad.
 Quadrant has peak. BUT not applied to the matrix.
 Pointers of peak all on border.

Function	Name
5	Constant
$\log \log(n)$	double log
$\log(n)$	logarithmic
$\log^2(n)$	Polylogarithmic
n	linear
$n \log(n)$	log-linear
n^3	polynomial
$n^3 \log(n)$	
n^4	polynomial
2^n	exponential
2^{2n}	
$n!$	factorial

Let $T(n) = O(f(n))$
 $S(n) = O(g(n))$
 $\Rightarrow T(n) + S(n) = O(f(n) + g(n))$
 $\Rightarrow T(n) S(n) = O(f(n) g(n))$

homogeneous soln.
 $T(n) = a_h + a_p$
 bounded by initials.
 Substitute & solve for a_p

$f(n)$	Trivial
Polynomial of degree n	Polynomial of degree n .
$(k+1)k \cdot c^n$	$A n c^n$
c^n	$A c^n$

AKRA BAZZI THEOREM.
 $T(n) = g(n) + \sum_{i=1}^k a_i T(b_i n + h_i(n))$

$T(n) = a T(n/b) + O(n^k)$ - $a, b > 0 - k \geq 0$.
 If $a < 1$, $T(n) = O(n^k)$
 If $a > 1$, $T(n) = O(n^{k+1})$
 If $a = 1$, $T(n) = O(n^{k+1})$
 $T(n) = O(a^n / b^n)$
 $T(n) = a T(n/b) + \theta(n^k \log^p n)$

Guess Particular Soln. of recurrence
LINEAR NONHOMOGENEOUS Recurrences.

Conditions:
 - Sufficient base cases.
 - $a_i > 0 \wedge 0 < b_i < 1$
 - $|g'(x)| = O(x^c)$
 - $|h_i(x)| = O(\frac{x}{(\log x)^r})$

$\Rightarrow$ Find $p: \sum_{i=1}^k a_i b_i^p = 1$
 $T(n) = \theta(n^p (1 + \int_1^n \frac{g(u)}{u^{p+1}} du))$
 $- a \geq 1 - b > 1 - k \geq 0 - p \in \mathbb{R} - c = \log_b a$
 If $a > b^k$, $T(n) = n^c$
 If $a = b^k$, $T(n) = n^c \log n$
 If $a < b^k$, $T(n) = \theta(n^k)$

MASTER THEOREM If $a < b^k$, $T(n) = \begin{cases} \theta(n^c \log^{p+1} n), & p > -1 \\ \theta(n^c \log \log n), & p = -1 \\ \theta(n^c), & p < -1 \end{cases}$

$T(n) = \begin{cases} \theta(n^c \log^{p+1} n), & p > -1 \\ \theta(n^c \log \log n), & p = -1 \\ \theta(n^c), & p < -1 \end{cases}$

ITERATIVE